

$$= 1.2 \times 10^9 \text{ W}$$

			Total	A1 [2] [6]
6	(a)	When two (or more) waves meet (<i>not 'superpose' or 'interfere'</i>) resultant <u>displacement</u> is the sum of individual (displacements)	B1 M1 A1	[3]
	(b)	(i) any correct line through points of intersection of crests (ii) any correct line through intersections of a crest and a trough	B1 B1	[2]
	(c)	(i) $\lambda = ax/D$ OR $\lambda = a \sin \theta$ and $\theta = x/D$ $650 \times 10^{-9} = (a \times 0.70 \times 10^{-3})/1.2$ $a = 1.1 \times 10^{-3} \text{ m}$ (ii) 1 no change 2 brighter 3 no change (<i>accept stay/remain dark</i>)	C1 C1 A1 B1 B1 B1	[3] [3]
			Total	[11]
7	(a)	(i) $P = VI$ current = $60/240 = 0.25 \text{ A}$	C1 A1	